

Heat engine

X21 (2021) — English translation

The Carnot heat engine operates with two heat reservoirs of finite heat capacity $C = 500 \text{ J/K}$, which at the initial moment of time are at temperatures $T_h = 625 \text{ K}$ and $T_c = 298 \text{ K}$. The power developed by a heat engine decreases exponentially with time $P = P_0 e^{-kt}$, where $P_0 = 960 \text{ W}$ is the power at the initial moment of time, and t is the time elapsed since the initial moment.

A1	At what temperature of the thermal reservoirs will the heat engine stop working?	1.00
A2	How much work will the heat engine do during its operation?	1.00
A3	After what time will the power developed by the heat engine be halved?	1.00
A4	After what time will the efficiency of a heat engine decrease by half?	1.00
A5	How long will it take for the temperature difference between the heat reservoirs to be halved?	1.00

Source: <https://pho.rs/p/101>